

1. Fundamental Concepts in Cell Biology

1.1 Cell theory & modern view

- Cell is the smallest unit that can perform all life processes (metabolism, growth, response, reproduction).
- Modern additions:
 - o Energy flow occurs within cells (respiration, photosynthesis).
 - o Hereditary information (DNA → RNA → protein) is housed and expressed in cells.
 - o All cells share basic biochemical machinery (ATP, DNA, RNA, ribosomes, membranes).

1.2 Prokaryotic vs eukaryotic cells (mechanistic view)

Prokaryotes

- Genome: single circular DNA; often supercoiled; no histones (bacteria) or archaeal type histones.
- Compartmentation: no internal membrane systems; transcription and translation are coupled.
- Surface: cell wall (peptidoglycan in bacteria), outer membrane in Gram negative, capsules, flagella (flagellin), pili.
- Energy: plasma membrane houses electron transport chains.

Eukaryotes

- Genome: multiple linear chromosomes packaged with histones into chromatin; transcription and translation decoupled (nucleus vs cytoplasm).
- Compartmentation: nucleus and organelles with specific microenvironments → spatial separation of pathways.
- Cytoskeleton: actin, microtubules, intermediate filaments; basis for shape, division, organelle positioning, motility.
- Endomembrane system: ER, Golgi, lysosomes, endosomes, vesicles.

2. Plasma Membrane & Cell Surface

2.1 Plasma membrane structure

- Fluid mosaic model: phospholipid bilayer with embedded proteins, cholesterol (in animals), and carbohydrates on outer face.
- Lipids: phospholipids (amphipathic), glycolipids, cholesterol.
- Proteins: integral (span bilayer, e.g. transporters, receptors) and peripheral (bound to surface or cytoskeleton).
- Asymmetry: inner vs outer leaflet composition; key for signaling (e.g. PS externalization in apoptosis).

2.2 Functions

- Selective permeability: controls entry/exit of ions, nutrients, waste.
- Signal transduction: receptors for hormones, growth factors, neurotransmitters.
- Cell-cell recognition and adhesion: glycoproteins, glycolipids, CAMs (cadherins, integrins).

prokaryotes

From Greek pro- "before" and karyon "nut, kernel" (referring to the nucleus). Literally means "before-kernel" or "pre-nucleus," signifying cells lacking a true nucleus. The term was established in the 1960s to distinguish them from eukaryotes.

eukaryotes

From Greek eu- "good, well, true" and karyon "nut, kernel" (referring to the nucleus). Literally means "true-kernel" or "true-nucleus," signifying cells having a membrane-bound nucleus. It describes the more complex cells of animals, plants, fungi, and protists.

- Scaffold: anchors cytoskeleton and extracellular matrix (via integrins).

2.3 Membrane transport

- Passive transport:
 - Simple diffusion (gases, small hydrophobic molecules).
 - Facilitated diffusion (via channels or carriers; no ATP; down gradient).
- Active transport:
 - Primary (ATP driven; e.g. Na⁺/K⁺ ATPase).
 - Secondary (uses existing ion gradients).
- Vesicular transport:
 - Endocytosis (phagocytosis, pinocytosis, receptor mediated).
 - Exocytosis (regulated and constitutive secretion).

3. Cytoplasm, Cytoskeleton, Cell Wall, ECM

3.1 Cytoplasm & cytosol

- **Cytosol**: crowded solution of proteins, metabolites, ions; site of glycolysis, many biosynthetic reactions.
- Highly organized: microdomains, phase separation, cytoskeletal tracks.

3.2 Cytoskeleton

Microfilaments (actin)

- Structure: two stranded helical polymers; polar; often anchored to membrane.
- Functions: cell shape, cortical tension, lamellipodia/filopodia, muscle contraction (with myosin), cytokinesis (contractile ring).

Microtubules

- Structure: hollow tubes of α/β tubulin dimers; plus and minus ends; dynamic instability.
- Functions:
 - Mitotic spindle; **chromosome** segregation.
 - Organelle positioning and vesicle transport (kinesins toward + end, dyneins toward - end).
 - Axoneme of cilia/flagella (9+2 arrangement).

Intermediate filaments

- Rope like fibers; no polarity; cell type specific (keratins, vimentin, neurofilaments, lamins).
- Mechanical strength; nuclear lamina; **desmosomes** and hemidesmosomes.

3.3 Cell wall (plants, fungi, bacteria)

cytosol

From Greek kytos "a hollow vessel" (cell) and Latin solvere "to loosen or dissolve" (via "-sol"). It is the liquid part of the cytoplasm, essentially the "cell-solution" or "cell-dissolved" substance. The term was coined to differentiate the liquid from the organelles suspended within it.

chromosome

From Greek chrōma "color" and sōma "body." Literally translates to "color-body," due to their strong affinity for staining dyes used in microscopy. The term was introduced in the late 19th century by Heinrich von Waldeyer-Hartz.

desmosomes

From Greek desmos "bond, tie" and sōma "body." Literally means "bonding-body," referring to their function as spot welds or anchors between cells. They are specialized structures that provide strong adhesion and mechanical stability to tissues.

Plants

- Composition: cellulose microfibrils in matrix of hemicellulose, pectin, proteins; often lignin (secondary wall).
- Functions: tensile strength, protection, turgor maintenance, cell shape, defense.

Fungi: chitin, glucans, mannoproteins.

Bacteria: peptidoglycan; Gram positive vs Gram negative architecture.

3.4 Extracellular matrix (ECM) in animals

- Components: fibrous proteins (collagen, elastin), glycoproteins (fibronectin, laminin), proteoglycans.
- Functions: mechanical support, substrate for cell migration, reservoir of growth factors, signaling via integrins (outside in and inside out).

4. Nucleus and Genetic Organization

4.1 Nuclear envelope & nuclear pores

- Double membrane: outer continuous with rough ER; perinuclear space.
- Nuclear pore complexes (NPCs): large macromolecular assemblies; regulate nucleocytoplasmic transport via signal sequences (NLS, NES) and transport receptors (importins/exportins, Ran GTP).

4.2 Chromatin organization

- DNA wound around histone octamers → nucleosomes (beads on a string).
- Higher order folding → euchromatin (less condensed, transcriptionally active) vs heterochromatin (condensed, inactive or structural).
- Epigenetic regulation: histone modifications, DNA methylation, chromatin remodeling complexes.

4.3 Nucleolus

- Not membrane bound; ribosomal RNA synthesis (by RNA polymerase I) and assembly of ribosomal subunits.
- Has fibrillar centers, dense fibrillar component, granular component.

4.4 Nuclear functions

- Storage and expression of genetic information; regulation of transcription.
- Coordination of DNA replication, repair, recombination.
- Cell cycle regulation via checkpoint machinery localized to nuclear structures.

5. Endomembrane System

5.1 Endoplasmic reticulum (ER)

Rough ER (RER)

- Ribosomes bound via receptors; site of synthesis of:
 - o Secreted proteins, membrane proteins, and proteins of ER, Golgi, lysosomes.
- Co translational translocation via signal peptide, SRP, translocon.
- Folding and quality control; N linked glycosylation; disulfide bond formation (protein

chromatin

From Greek chrōma "color" (referring to staining) and the suffix -in "substance." Literally translates to "color-substance," the material that makes up chromosomes, due to its staining properties. It is the complex of DNA and proteins (histones) found inside the eukaryotic nucleus.

ribosomes

A blend of ribose (a sugar component of RNA) and Greek sōma "body." It is the "RNA-body," a molecular machine essential for protein synthesis (translation). The term highlights their composition of ribosomal RNA (rRNA) and protein.

disulfide isomerase).

- ER stress and unfolded protein response (UPR) when folding capacity is overwhelmed.

Smooth ER (SER)

- Lipid and steroid synthesis, membrane biogenesis.
- Detoxification (cytochrome P450 enzymes) in liver.
- Calcium storage and release (sarcoplasmic reticulum in muscle).

5.2 Ribosomes

- Ribonucleoprotein complexes (large + small subunits).
- Cytosolic (free) ribosomes: proteins for cytosol, nucleus, mitochondria, peroxisomes.
- Membrane bound (on RER): proteins for secretion, membranes, endomembrane organelles.

5.3 Golgi apparatus

- Flattened cisternae: cis Golgi (toward ER), medial, trans Golgi network.
- Functions:
 - o Modifies N linked oligosaccharides, adds O linked glycosylation.
 - o Sorts proteins based on signal motifs into different vesicles (secretory, lysosomal, plasma membrane).
 - o Produces lysosomal enzymes tagged with mannose 6 phosphate.
- Key for polarized trafficking and cell surface composition.

5.4 Lysosomes

- Acidic organelles (pH ~5) with hydrolytic enzymes (proteases, nucleases, glycosidases, lipases, phosphatases).
- Pathways:
 - o Endocytosis → early endosome → late endosome → lysosome.
 - o Autophagy: macroautophagy (double membrane autophagosome), microautophagy, chaperone mediated autophagy.
- Roles: degradation and recycling of macromolecules, organelle turnover, response to starvation.
- Lysosomal storage diseases: defects in specific enzymes → accumulation of substrates.

5.5 Endosomes

- Early endosomes: sorting station; decide between recycling to membrane vs degradation.
- Late endosomes/multivesicular bodies: intermediates before lysosomes.
- Crucial in receptor downregulation, signaling termination or propagation.

6. Peroxisomes, Vacuoles, Storage Organelles

6.1 Peroxisomes (microbodies)

- Single membrane organelles; contain oxidases and catalase.
- Functions:
 - o β oxidation of very long chain fatty acids (in animals).

lysosomes

From Greek lysis "to loosen, dissolve" and sōma "body." Literally means "dissolving-body," referring to their role in breaking down waste materials and cellular debris. They contain hydrolytic enzymes and function as the cell's waste disposal system.

peroxisomes

A blend of peroxide (referring to hydrogen peroxide) and Greek sōma "body." Literally the "peroxide-body," as they are the site of reactions that generate and then safely break down hydrogen peroxide. They are tiny organelles involved in various metabolic processes, including fatty acid breakdown.

- o Detoxification of hydrogen peroxide and other reactive species.
- o In plants: glyoxysomes convert lipids to carbohydrates in germinating seeds.
- Assembly: import of matrix proteins with PTS signals from cytosol; growth and division.

6.2 Vacuoles (mostly in plants, fungi)

- Large central vacuole in mature plant cells; can occupy most of cell volume.
- Functions:
 - o Storage (ions, metabolites, pigments, defensive compounds).
 - o Osmoregulation and turgor maintenance.
 - o Degradative functions (lysosome like).
- Tonoplast: vacuolar membrane with specific transporters.

6.3 Storage granules & inclusions

- Glycogen granules, lipid droplets, pigment granules.
- Not membrane bound (e.g. glycogen) or monolayer bound (lipid droplets).
- Energy reserves, thermal insulation, signaling platform (lipid droplets).

7. Mitochondria & Chloroplasts (Endosymbiotic Organelles)

7.1 Mitochondria

- Double membrane: outer smooth; inner folded into cristae; intermembrane space; matrix.
- Own circular DNA, ribosomes (bacterial like), independent replication (binary fission).
- Functions:
 - o Oxidative phosphorylation (ETC in inner membrane; ATP synthase).
 - o Citric acid cycle (matrix), fatty acid β oxidation, amino acid metabolism.
 - o Apoptosis regulation: cytochrome c release; intrinsic pathway.
 - o Ca^{2+} buffering; ROS generation and detoxification.

7.2 Chloroplasts (and other plastids in plants)

- Double membrane; internal thylakoid membranes stacked into grana; stroma.
- Chlorophyll and other pigments in thylakoid membranes.
- Light reactions: electron transport, photophosphorylation, NADPH formation.
- Dark reactions (Calvin cycle) in stroma: CO_2 fixation $\rightarrow$ sugars.
- Plastids differentiation: chloroplasts, chromoplasts (pigments), amyloplasts (starch storage).

7.3 Endosymbiotic theory

- Mitochondria and chloroplasts arose from engulfed bacteria (α proteobacteria and cyanobacteria).
- Evidence: double membranes, circular DNA, bacterial type ribosomes, sensitivity to some antibiotics, division by fission.

8. Centrosomes, Cilia, Flagella, Microvilli

8.1 Centrosomes & centrioles

- Centrosome: main microtubule organizing center (MTOC) in animal cells; near nucleus.
- Contains a pair of centrioles (barrel shaped microtubule triplets) + pericentriolar material.
- Functions: nucleate and anchor microtubules; organize mitotic spindle.

8.2 Cilia and flagella (eukaryotic)

- Axoneme: 9 doublet microtubules + 2 central microtubules ("9+2").
- Powered by dynein motor proteins; bending via sliding of microtubule doublets.
- Primary (non motile) cilia: sensory roles; 9+0 structure.

mitochondria

From Greek mitos "thread" and chondrion "small grain" or "granule." Literally means "thread-granules," a name derived from their thread-like or granular appearance under the microscope. They are the "powerhouses" of the cell, generating most of the cell's supply of ATP.

- Functions: locomotion (single celled eukaryotes), fluid movement across epithelia (respiratory tract, oviduct), mechanosensation.

8.3 Microvilli & other projections

- Microvilli: actin based projections increasing surface area (intestinal epithelium, kidney tubules).
- Stereocilia: long microvillus like structures (epididymis, inner ear hair cells).
- Filopodia/lamellipodia: dynamic actin structures for migration and environmental sensing.

9. Cell Junctions and Communication (brief)

- Tight junctions (occluding): seal cells; maintain polarity and barrier (epithelia).
- Adherens junctions & desmosomes (anchoring): link cytoskeletons of neighboring cells (actin or intermediate filaments).
- Hemidesmosomes: anchor cells to ECM.
- Gap junctions: connexon channels; allow ion/small molecule exchange; electrical and metabolic coupling.
- Plasmodesmata (plants): cytoplasmic channels through cell walls.

10. Cell Cycle, Division & Control

10.1 Cell cycle phases

- G₁: growth, biosynthesis, preparation for DNA replication; decision point (restriction point).
- S: DNA replication; histone synthesis; centrosome duplication.
- G₂: further growth, preparation for mitosis (check DNA integrity).
- M phase: mitosis (nuclear division) + cytokinesis (cytoplasm division).
- G₀: quiescent state; reversible (e.g. liver) or terminally differentiated (neurons, muscle).

10.2 Mitosis (mechanistic view)

- Prophase: chromatin condensation; centrosomes move apart; spindle begins to form; nuclear envelope intact.
- Prometaphase: nuclear envelope breakdown; spindle microtubules attach to kinetochores.
- Metaphase: chromosomes aligned at metaphase plate; bipolar attachment under tension.
- Anaphase: cohesion cleaved; sister chromatids segregate to opposite poles; spindle elongates.
- Telophase: chromosomes decondense; nuclear envelopes reassemble; contractile ring constricts.
- Cytokinesis: actomyosin ring divides cell into two.

10.3 Meiosis (very short, for contrast)

- Two successive divisions after one S phase → 4 haploid cells.
- Meiosis I: homologous chromosomes pair and segregate (reductional).
- Meiosis II: sister chromatids segregate (equational).
- Crossing over (prophase I) and independent assortment generate genetic diversity.

10.4 Molecular control: cyclins, CDKs, checkpoints

- CDKs: serine/threonine kinases; activated by binding specific cyclins and by phosphorylation state.
- Cyclin levels oscillate; CDK levels constant.
- Major checkpoints:
 - o G₁/S: DNA damage? size? environment (growth factors, nutrients)?
 - o G₂/M: DNA fully replicated undamaged?
 - o Spindle assembly checkpoint: all kinetochores attached and under proper tension?

Key regulators:

- p53: DNA damage sensor; induces cell cycle arrest (via p21), DNA repair, or apoptosis.
- Rb: controls G₁/S transition by binding E2F; phosphorylation by cyclin-CDKs releases E2F →

S phase gene expression.

- Oncogenes (e.g. Ras, Myc) and tumor suppressor genes (p53, Rb) integrate growth signals and stress.

11. Apoptosis & Cell Turnover (bonus but high yield at graduate level)

- Programmed cell death; characterized by cell shrinkage, chromatin condensation, membrane blebbing, apoptotic bodies; no inflammation.
- Pathways:
 - o Intrinsic (mitochondrial): Bcl 2 family, cytochrome c, caspase 9.
 - o Extrinsic (death receptor): Fas, TNF receptor, caspase 8.
- Maintains tissue homeostasis; removes damaged or potentially oncogenic cells.